SuperStore Dashboard: Step-by-Step Rebuild Guide

1. Project architecture

Excel file → Power BI Get Data → Power Query data typing/cleaning → data model → DAX measures → report visuals → slicer and interactions → validation → publish.

2. Import the dataset

- Open Power BI Desktop > Home > Get data > Excel workbook.
- Select SuperStore_Sales_Dataset.xlsx and the data sheet; choose Transform Data.
- Confirm headers and set Order Date/Ship Date to Date, Sales/Profit to Decimal Number, Quantity to Whole Number.
- Remove or rename the malformed first header; inspect ind1/ind2 and remove only if unused.
- Normalize Returns values (#N/A to blank/unknown) and audit before Close & Apply.
- Rename the table to SuperStore.

3. Data validation and grain

- One Excel row represents an order line; do not equate row count to order count.
- Check nulls, duplicate complete rows, negative Sales/Profit, and Ship Date before Order Date.
- Distinct order IDs are the correct base for an order count.
- If an order has multiple lines, averaging delivery days by row weights orders with more lines more heavily.

4. Create DAX calculated columns

In Data/Model view, select New column:

Delivery Days = DATEDIFF (SuperStore[Order Date], SuperStore[Ship Date], DAY)

Order Year = YEAR (SuperStore[Order Date])

Order Month No = MONTH (SuperStore[Order Date])

Order Month = FORMAT (SuperStore[Order Date], "MMMM")

Sort Order Month by Order Month No. Exclude negative delivery days or resolve data errors before aggregating.

5. Create DAX measures

Modeling > New measure. Create each separately:

Total Sales = SUM (SuperStore[Sales])

Total Profit = SUM (SuperStore[Profit])

Total Quantity = SUM (SuperStore[Quantity])

Total Orders = DISTINCTCOUNT (SuperStore[Order ID])

Profit Margin % = DIVIDE ([Total Profit], [Total Sales])

Avg Delivery Days (Line Weighted) = AVERAGEX (FILTER (SuperStore, SuperStore[Delivery Days] >= 0), SuperStore[Delivery Days])

For a true order-weighted metric, calculate a delivery day per distinct Order ID using a documented rule; validate whether order lines share the same dates.

6. KPI cards

- Insert > Card; add Total Sales, Total Orders, Total Profit and Avg Delivery Days.
- Set display units to Auto/Millions/Thousands as appropriate; adjust decimals.
- Apply purple panel background, cyan numeric text, rounded corners and consistent typography.

7. Monthly trend charts

- Create a stacked area chart for Sales by Month and Year: X-axis Order Month; Legend Order Year; Y-axis Total Sales.
- Create a second stacked area chart for Profit by Month and Year: Y-axis Total Profit.
- Sort month names by Order Month No; check chart interpretation because stacked area charts aggregate series vertically.

8. Doughnut and bar charts

- Doughnut: Legend Payment Mode; Values Total Sales. Repeat for Region and Segment.
- Clustered bar: Ship Mode vs Total Sales; Category vs Total Sales; Sub-Category vs Total Sales.
- Sort by Total Sales descending; show data labels and set display units to millions.

9. Geographic map

- Insert Map visual; Location = State; Size = Total Sales; optionally tooltips = Total Profit and Total Orders.
- Set State data category to State or Province and use Country as additional geographic context.
- Validate geocoding and avoid interpreting bubble size as profit if Size uses Sales.

10. Region slicer and connections

- Insert Slicer visual and drag Region to Field.
- Format > Slicer settings > Style: Tile (where available); orient horizontally and use four items.
- Format ribbon > Edit interactions; click slicer and set each chart to Filter, not None, where appropriate.
- Test Central/East/South/West one at a time; verify KPI totals and map/chart changes.
- If creating additional report pages, View > Sync slicers to synchronize region selections across pages.

11. Full-screen and refresh in Power BI

- Full-screen reading: use Power BI Service full-screen mode after publishing, or Power BI Desktop View > Page view > Fit to page (not true Windows fullscreen).
- Refresh: Home > Refresh in Power BI Desktop. In Power BI Service, configure credentials/gateway and scheduled refresh if applicable.
- For clickable report navigation, use Insert > Buttons and assign Action = Page navigation or Bookmark.

- Power BI does not execute VBA; do not assign Excel macros to PBIX buttons.

12. Optional Excel VBA controls (separate Excel workbook)

If you also build an Excel dashboard, save as .xlsm, press Alt+F11 > Insert > Module and paste:

```
Sub ToggleFullScreen()
    Application.DisplayFullScreen = Not Application.DisplayFullScreen
End Sub

Sub RefreshDashboard()
    ThisWorkbook.RefreshAll
End Sub
```

On an Excel sheet: Insert > Shapes > right-click > Assign Macro. These are reproducible example macros, not extracted from the PBIX.

13. Validation and publication

- Compare source sums and distinct counts to DAX measures with all filters cleared.
- Inspect Ship Date/Order Date anomalies and document exclusions.
- Confirm slicer interaction with every visual.
- Check tooltips, color contrast, labels, and mobile readability.
- Save PBIX, export screenshot, add README and PDF; publish only data you have permission to share.